

Micro I, class 13

We began by working through MWG 10.C.3, which asserts that the *aggregate marginal cost function* (from the ACM problem) is the economy's *aggregate inverse supply*: $S^{-1}(Q) = C'(Q)$. An *identical* argument (save for notation) shows that the aggregate marginal utility function (from problem H) equals the economy's aggregate inverse demand: $D^{-1}(Q) = \Phi'(Q)$. I don't think that I emphasized sufficiently that, at a competitive equilibrium, the allocation of good-1 outputs among firms satisfies "aggregate cost minimization;" and the allocation of good 1 among consumers satisfies "aggregate utility maximization." *Firms and consumers are lead as if by an invisible hand to promote an end which is no part of their intention.*

I repeat that the two-good quasilinear economy satisfies the Gorman conditions for a positive (and normative) representative consumer. That consumer has preferences represented by $U(X, M) = \Phi(X) + M$, where $\Phi(\cdot)$ is the value function for problem (AUM) in my notes (the same thing as the function $\gamma(\cdot)$ in MWG 10.C.4).

The main event after that is the issue of *existence* of a competitive equilibrium in the two-good quasilinear economy. We defined aggregate excess demand to be $Z(p) = D(p) - S(p)$. If $0 \in Z(p^*)$, then p^* is a competitive equilibrium price. (The allocation of good 1 can be recovered from the individual demands and supplies, and of good 2 from the budget constraints and cost functions.). For $L = 2$, existence follows from an application of the *intermediate value theorem* from calculus, a special kind of a *fixed-point theorem*. I first gave sufficient conditions directly on the excess demand: it is single-valued and continuous, with two prices p_0 and p_1 with $Z(p_0) \geq 0 \geq Z(p_1)$. Between p_0 and p_1 , then, there must be a price p^* so that Z takes on the intermediate value of 0. I hope that I convinced you that these conditions in turn are satisfied if each ϕ_i is C^1 , strictly increasing, and strictly concave with $\lim_{x \rightarrow \infty} \phi'_i(x) = 0$; and if each c_j is C^1 , strictly increasing, and strictly convex with $\lim_{q \rightarrow \infty} c'_j(q) = \infty$; and each $f_j(0) = 0$. *These cost conditions preclude nondecreasing returns to scale and in particular **increasing** returns to scale ($f_j(\lambda z) > \lambda f_j(z)$ for every $z > 0$ and $\lambda > 1$).*

I mentioned that nonconvexities in preferences can *also* lead to equilibrium nonexistence. I then vaguely mentioned that Aumann (1966) showed that, with a *continuum* of consumers, a competitive equilibrium exists, even if consumer preferences are not convex.

Since a continuum of consumers better captures the intuitive idea that individuals under perfect competition are *negligible*, I suggested that the main economically-substantive reason why a competitive equilibrium might not exist is *increasing returns to scale*.